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Rangkuman Minggu 8 Mikroprosesor dan Interface

LEA SI, DAT

Op code						D	W	Mod		Reg			R/M		
1	0	0	0	1	1	0	1	0	1	1	1	0	1	1	0
8				D				7				6			

MOV BX,1234 (alternate encoding)

Op code				W	reg			LD/DLB	HD/DHB
1	0	1	1	1	0	1	1	34(dec)	12(dec)
B				B				D2	04

MOV CX, 1234H

Op code				W	reg			LD/DLB	HD/DHB
1	0	1	1	1	0	0	1	34	12
B				9				34	12

MOV DX,[1000]

Op code						D	W	Mod		Reg			R/M			LD/DLB	HD/DHB
1	0	0	0	1	0	1	1	0	0	0	1	0	1	1	0	00(dec)	10(dec)
8				B				1				6				E8	03

MOV SI,[1000H]

Op code						D	W	Mod		Reg			R/M			LD/DLB	HD/DHB
1	0	0	0	1	0	1	1	0	0	0	1	0	1	1	0	00	10
8				B				1				6				00	10

MOV AX,[BX]

Op code						D	W	Mod		Reg			R/M		
1	0	0	0	1	0	1	1	0	0	0	0	0	1	1	1
8				B				0				7			

MOV AX, BX

Op code						D	W			Reg1			Reg2		
1	0	0	0	1	0	1	1	1	1	0	0	0	0	1	1
8				B				C				3			

MOV DX, [BX+SI]

Op code						D	W	Mod		Reg			R/M		
1	0	0	0	1	0	1	1	0	0	0	1	0	0	0	0
8				B				1				0			

R/M[BX+SI]

Mod[BX+SI]

MOV [BP+SI], BL

Op code						D	W	Mod		Reg			R/M		
1	0	0	0	1	0	0	0	0	0	0	1	1	0	1	0
8				8				1				A			

R/M[BP+SI]

Mod[BP+SI]

MOV [BP+DI-123H], CH

Op code						D	W	Mod		Reg			R/M			LD	HD
1	0	0	0	1	0	0	0	1	0	1	0	1	0	1	1	23	-01
8				8				A				B				DD	FE

R/M[BP+DI]

Mod[BP+DI-123H]

MOV DX, [BX+SI+1122H]

Op code						D	W	Mod		Reg			R/M			LD	HD
1	0	0	0	1	0	1	1	1	0	0	1	0	0	0	0	22	11
8				B				9				0				22	11

R/M[BX+SI]

Mod[BX+SI+1122H]

MOV DX, [BX+SI+1122]

Op code						D	W	Mod		Reg			R/M			LD	HD
1	0	0	0	1	0	1	1	1	0	0	1	0	0	0	0	22(dec)	11(dec)
8				B				9				0				62	04

R/M[BX+SI]

Mod[BX+SI+1122]

MOV W.[BX+DI+1794H], 1234H

Op code						D	W	Mod					R/M			LD	HD	DLB	DHB
1	1	0	0	0	1	1	1	1	0	0	0	0	0	0	1	94	17	34	12
C				7				8				1				94	17	34	12

R/M[BX+DI]

Mod[BX+DI+1794H]

MOV W.[BX+DI+1000], 14H

Op code						D	W	Mod					R/M			LD	HD	
1	1	0	0	0	1	1	1	1	0	0	0	0	0	0	1	00(dec)	10(dec)	14
C				7				8					1			E8(hex)	03(hex)	14

R/M[BX+DI]

Mod[BX+DI+1000]

MOV [BX+DI+1000], 14H

Op code						D	W	Mod					R/M			LD	HD	
1	1	0	0	0	1	1	0	1	0	0	0	0	0	0	1	00(dec)	10(dec)	14
C				6				8					1			E8(hex)	03(hex)	14

R/M[BX+DI]

Mod[BX+DI+1000]

MOV W.[BX+DI+1056H], 14H

Op code						D	W	Mod					R/M			LD	HD	DLB	DHB
1	1	0	0	0	1	1	1	1	0	0	0	0	0	0	1	56	10	14	00
C				7				8					1			56	10	14	00

R/M[BX+DI]

Mod[BX+DI+1056H]

MOV [BX+SI+1234H],5678H

Op code						D	W	Mod					R/M			LD	HD	DLB	DHB
1	1	0	0	0	1	1	1	1	0	0	0	0	0	0	0	34	12	78	56
C				7				8					0			34	12	78	56

R/M[BX+SI]

Mod[BX+SI+1234H]

MOV ES:[BP+DI+1E3DH], 1234H

Segment overide	Op code						D	W	Mod					R/M			LD	HD	DLB	DHB
ES	1	1	0	0	0	1	1	1	1	0	0	0	0	0	1	1	3D	1E	34	12
26	C				7				8					3			3D	1E	34	12

R/M[BP+DI]

Mod[BP+DI+1EDH]

MOV SS:[BX+DI+10], 1234H

Segment overide	Op code						D	W	Mod					R/M			LD	HD	DLB	DHB
SS	1	1	0	0	0	1	1	1	0	1	0	0	0	0	0	1	0	10(dec)	34	12
36	C				7				4					1			0	A(hex)	34	12

R/M[BX+DI]

Mod [BX+DI+10]

MOV CS:[BX+SI+6723H],7809H

Segment overide	Op code						D	W	Mod					R/M			LD	HD	DLB	DHB
CS	1	1	0	0	0	1	1	1	1	0	0	0	0	0	0	0	23	67	09	78
2E	C				7				8					4			23	67	09	78

R/M [BX+SI]

Mod[BX+SI+6723]

MOV DS:[SI+4321H],3456H

Segment overide	Op code						D	W	Mod					R/M			LD	HD	DLB	DHB
DS	1	1	0	0	0	1	1	1	1	0	0	0	0	1	0	0	21	43	56	34
3E	C				7				8					4			21	43	56	34

R/M [SI]

Mod[SI+4321H]

ADD AL,88

Op code						D	W	Immediate data
0	0	0	0	0	1	0	0	88 (dec)
0				4				Ke heksa 58

DAA

Op code							
0	0	1	0	0	1	1	1
2				7			

ADD AL,88H

Op code						D	W	Immediate data
0	0	0	0	0	1	0	0	88H
0				4				88

ADC AH,CL

Op code						D	W			Reg1		Reg2			
0	0	0	1	0	0	1	0	1	1	1	0	0	0	0	1
1				2				E				1			

AND CL,AL

Op code						D	W			Reg1			Reg 2		
0	0	1	0	0	0	1	0	1	1	0	0	1	0	0	0
2				2				C				8			

OR BX,DX

Op code						D	W			Reg1			Reg 1		
0	0	0	0	1	0	1	1	1	1	0	1	1	0	1	0
0				B				D				A			

XOR [1234H],DX

Op code						D	W	Mod		Reg			R/M			LD/DLB	HD/DHB
0	0	1	1	0	0	0	1	0	0	0	1	0	1	1	0	34	12
3				1				1				6				34	12

SHL AX,1

Op code						D	W					Reg			
1	1	0	1	0	0	0	1	1	1	1	0	0	0	0	0
D				1				E				0			

SHR AX,CL

Op code						D	W					Reg			
1	1	0	1	0	0	1	1	1	1	1	0	1	0	0	0
D				3				E				8			

ROR CL,1

Op code						D	W						Reg		
1	1	0	1	0	0	0	0	1	1	0	0	1	0	0	1
D				0				C				9			

```
TITLE PERKALIAN 8 BIT DENGAN 8 BIT
.MODEL SMALL   ;CH*CL=DX
.STACK 100H
.DATA
.CODE
MAIN PROC
     MOV AX,@DATA
     MOV DS,AX

     MOV DX,0
     ADD CL,0
     JZ  SKIP

LUP: ADD DL,CH
     ADC DH,0
     DEC CL
     JNZ LUP
SKIP:
     MOV AH,4CH
     INT 21H
     MAIN ENDP
END MAIN
```

```
TITLE 03_PEMBAGIAN 16 BIT DENGAN 8 BIT
.MODEL SMALL   ;BL/CL=DX
.STACK 100H
.DATA
.CODE
MAIN PROC
     MOV AX,@DATA
     MOV DS,AX

     MOV DX,0
     ADD BX,0
     JZ  SKIP
     ADD CL,0
     JZ  SKIP

LUP: SUB BL,CL
     SBB BH,0
     ADD DX,1
     CMP BH,0
     JA  LUP
     CMP BL,CL
     JAE LUP
     MOV AL,BL
SKIP:
     MOV AH,4CH
     INT 21H
     MAIN ENDP
END MAIN
```

```
TITLE 04_PEMBAGIAN 32 BIT DENGAN 16 BIT
.MODEL SMALL
.STACK 100H
.DATA
HASILBAGI DW 0,0
SISABAGI DW 0
.CODE
MAIN PROC
     MOV AX,@DATA
     MOV DS,AX

     MOV DX,0
     MOV AX,BX
     ADD AX,CX
     JZ  SKIP
     ADD SI,0
     JZ  SKIP
     MOV AX,0

LUP: SUB CX,SI
     SBB BX,0
     ADD DX,1
     ADC AX,0
     CMP BX,0
     JA  LUP
     CMP CX,SI
     JAE LUP
     MOV [SISABAGI],CX
     MOV [HASILBAGI],DX
     MOV [HASILBAGI+2],AX
SKIP:
     MOV AH,4CH
     INT 21H
     MAIN ENDP
END MAIN
```

```
TITLE 05_PERKALIAN 16 BIT DENGAN 8 BIT
.MODEL SMALL
.STACK 100H
.DATA
OP1 DW 12E2H
OP2 DB 3CH
HASIL DW 0
.CODE
MAIN PROC
     MOV AX,@DATA
     MOV DS,AX

     MOV CX,0
     MOV BX,[OP1]
     MOV CL,[OP2]
     MOV DX,0
     ADD BX,0
     JZ  SKIP
     ADD CL,0
     JZ  SKIP

LUP: ADD DX,BX
     ADC CH,0
     DEC CL
     JNZ LUP
     MOV CL,CH
     MOV CH,0
     MOV [HASIL],DX
     MOV [HASIL+2],CX
SKIP:
     MOV AH,4CH
     INT 21H
     MAIN ENDP
END MAIN
```